

A NOTE ON

A toy model of mortgage market frictions

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1. Overview

In a pathbreaking contribution, [Kiyotaki and Moore \(1997\)](#) introduced a collateral constraint into a standard dynamic macroeconomic model. On financial markets, the collateral acts as security pledged for the payment of a loan. For instance, a firms borrowing loans to finance working capital may give to banks some stocks and bonds as collateral. In case of firms default, bank may seize the collateral to repay the loan. While [Kiyotaki and Moore \(1997\)](#)'s model was a simple toy model featuring a farmer using her land as collateral, [Iacoviello \(2005\)](#) tailored collateral constraint to housing purchase and physical capital renting, and estimated his model by matching his model's IRFs on those of a VAR model. The financial crisis, which occurred two years after the publication of his paper in the AER, made his model very popular to explain disruptions in financial markets through unexpected shocks on the *ex ante* value of collateral.¹

Later, [Gerali et al. \(2010\)](#) provided an extension of [Iacoviello \(2005\)](#) by introducing imperfect competition and rate adjustment for banks. In their setup, bank capital became key in explaining business cycles of the Euro Area. Since [Guerrieri and Iacoviello \(2017\)](#), it is now possible to estimate macroeconomic models with occasionally binding constraints. Prior to this contribution, collateral are eternally binding as they always affect the borrowers discounting in their Euler equation. However, during favorable economics periods as those prior to the financial crisis, collateral requirement were less important suggesting that collateral constraint was loosening, or simply not binding. [Guerrieri and Iacoviello \(2015\)](#) tackled this problem of eternally binding constraint by introducing a sort of markov-switching mechanism allowing any constraint to bind occasionally, referred to as a *piece-wise* solution. However to estimate a model, we usually employ the Kalman filter for unobservable variable to construct the likelihood function, the later is linear and thus not meant for the case of occasionally binding constraint. [Guerrieri and Iacoviello \(2017\)](#) solve this problem by using an *inversion filter*.

¹For extensions, see [Angelini et al. \(2011\)](#) for an extension to macroprudential policy analysis, [Iacoviello and Neri \(2010\)](#) for trends.

2. The model

I develop here a toy model of credit market for firms that relies on the collateral constraint mechanism *à la* Kiyotaki and Moore (1997).

2.1 Patient Households

Suppose a welfare maximizing household enjoying current and future consumption and disliking working with respective curvatures $\sigma_C, \sigma_H > 0$. The household maximizes the following stream of utilities discounted at a fixed rate β_P :

$$\max_{\{C_t^P, d_t, H_t^P, L_t^P\}} \sum_{\tau=0}^{\infty} \beta_P^\tau E_t \left\{ \varepsilon_{t+\tau}^C \log(C_{t+\tau}^P) + j \log(H_{t+\tau}^P) - \varepsilon_{t+\tau}^H \chi_P \frac{(L_{t+\tau}^P)^{1+\sigma_H}}{1+\sigma_H} \right\} \quad (1)$$

where C_t^P is consumption, H_t^P is the housing stock with utility parameter $j \geq 0$, L_t is hours worked and $\chi_P > 0$ is a shift parameter allowing to pin down the amount of hours worked in steady state. ε_t^H denotes an AR(1) labor supply shock as in Smets and Wouters (2003) while ε_t^C is a preference shock on consumption.

The budget constraint in real terms reads as:

$$C_t^P + q_t (H_t^P - (1 - \delta) H_{t-1}^P) + d_t = \frac{R_{t-1}}{\pi_t} d_{t-1} + L_t^P w_t^P - T_t, \quad (2)$$

where q_t is the price of housing, δ is the depreciation housing, d_t is the amount of real deposits, R_{t-1} is the pre-determined nominal interest rate of real deposits d_t , w_t the real wage is computed as $w_t^P = W_t^P / P_t$ and a lump-sum tax financing the government T_t .

The Lagrangian problem reads as:

$$\begin{aligned} \mathcal{L}_t^P = & \sum_{\tau=0}^{\infty} \beta_P^\tau \left[\varepsilon_{t+\tau}^C \log(C_{t+\tau}^P) + j \log(H_{t+\tau}^P) - \varepsilon_{t+\tau}^H \chi_P \frac{(L_{t+\tau}^P)^{1+\sigma_H}}{1+\sigma_H} \right] \\ & + \beta_P^\tau \lambda_{t+\tau}^P \left[R_{t-1+\tau} d_{t-1+\tau} + L_{t+\tau}^P w_{t+\tau}^P - T_{t+\tau} - C_{t+\tau}^P - q_{t+\tau} (H_{t+\tau}^P - (1 - \delta) H_{t-1+\tau}^P) - d_{t+\tau} \right] \end{aligned}$$

where λ_t^P is a lagrangian multiplier.

First order conditions are:

$$C_t^P : \varepsilon_t^C (C_t^P)^{-1} - \lambda_t^P = 0 \quad (3)$$

$$d_t : \beta_H E_t \{ \lambda_{t+1}^H \} R_t - \lambda_t^H = 0 \quad (4)$$

$$H_t^P : \frac{j}{H_t^P} - q_t \lambda_t^P + (1 - \delta) \beta_P q_{t+1} \lambda_{t+1}^P = 0 \quad (5)$$

$$L_t^P : -\varepsilon_t^H \chi_P (L_t^P)^{\sigma_H} + \lambda_t^P w_t^P = 0 \quad (6)$$

2.2 Impatient Households

Unlike patient households, impatient households are borrowing loans to finance their loans. They face

$$R_t^b b_t = m E_t \{ q_{t+1} \} H_t^P \quad (7)$$

where $m \in [0, 1]$ is the Loan-To-Value ratio (LTV, here after) which is fixed as this constraint is eternally binding. The value of the collateral is q_{t+1} . It is possible to add a shock to this equation to features exogenous shock to the borrowing capacity of households. As explained in [Gerali et al. \(2010\)](#), from a microeconomic point of view, $(1 - m)$ can be interpreted as the proportional cost of collateral repossession for banks in case default should occur. At a macro-level, the value of m determines the amount of credit that banks make available to households, for a given (discounted) value of their housing stock.

In a similar fashion as for patient households, impatient households are maximizing the following welfare index:

$$\max_{\{C_t^I, b_t, H_t^I, L_t^I\}} \sum_{\tau=0}^{\infty} \beta_I^\tau E_t \left\{ \varepsilon_{t+\tau}^C \log(C_{t+\tau}^I) + j \log(H_{t+\tau}^I) - \varepsilon_{t+\tau}^H \chi_I \frac{(L_{t+\tau}^I)^{1+\sigma_H}}{1 + \sigma_H} \right\} \quad (8)$$

where C_t^I is consumption, H_t^I is the housing stock with utility parameter $j \geq 0$, L_t^I is hours worked and $\chi_I > 0$ is a shift parameter allowing to pin down the amount of hours worked in steady state.

The budget constraint in real terms reads as:

$$C_t^I + q_t (H_t^I - (1 - \delta) H_{t-1}^I) + R_{t-1}^b b_{t-1} = b_t + L_t^I w_t^I, \quad (9)$$

where b_t is the amount of real loans, R_{t-1}^b is the pre-determined nominal borrowing rate and w_t^I the real wage is computed as $w_t^I = W_t^I / P_t$.

The Lagrangian problem reads as:

$$\begin{aligned} \mathcal{L}_t^I = & \sum_{\tau=0}^{\infty} \beta_I^\tau \left[\varepsilon_{t+\tau}^C \log(C_{t+\tau}^I) + j \log(H_{t+\tau}^I) - \varepsilon_{t+\tau}^H \chi_I \frac{(L_{t+\tau}^I)^{1+\sigma_H}}{1 + \sigma_H} \right] \\ & + \beta_I^\tau \lambda_{t+\tau}^I \left[b_{t+\tau} + L_{t+\tau}^I w_{t+\tau}^I - C_{t+\tau}^I - q_{t+\tau} \Delta (H_{t+\tau}^I - (1 - \delta) H_{t-1+\tau}^I) - R_{t-1+\tau}^b b_{t-1+\tau} \right] \\ & + \beta_I^\tau \lambda_{t+\tau}^I \varphi_{t+\tau}^I \left[b_{t+\tau} - m E_t \{q_{t+1+\tau}\} H_{t+\tau} / R_{t+\tau}^b \right] \end{aligned}$$

where λ_t^I and φ_t^I are a lagrangian multipliers associated to the budget and collateral constraints. Here, we simplify the problem for the borrowing constraint by pre-multiplying by λ_t^I to express each constraints into consumption equivalents.

First order conditions are:

$$C_t^I : (C_t^I)^{-1} - \lambda_t^I = 0 \quad (10)$$

$$b_t : \beta_I E_t \{ \lambda_{t+1}^I \} R_t^b - \lambda_t^I + \lambda_t^I \varphi_t^I = 0 \quad (11)$$

$$H_t^I : \frac{j_I}{H_t^I} - q_t \lambda_t^I + (1 - \delta) \beta_I q_{t+1} \lambda_{t+1}^I = 0 \quad (12)$$

$$L_t^I : -\chi_I (L_t^I)^{\sigma_H} + \lambda_t^I w_t^I = 0 \quad (13)$$

2.3 Firms

Let us now suppose that there is a large number of identical wholesale firms. They hire both type of workers. Firms technology is given by:

$$Y_t = \varepsilon_t^A (L_t^P)^\lambda (L_t^I)^{1-\lambda} \quad (14)$$

where ε_t^A is the a technology shock, L_t^P is the demand for patient hours worked with intensity in technology of λ and L_t^I is the same for impatient households.

The cost minimization problem is given by:

$$\begin{aligned} \min_{\{L_t^P, L_t^I\}} & w_t^P L_t^P + w_t^I L_t^I \\ \text{s.t. } & Y_t = \varepsilon_t^A (L_t^P)^\lambda (L_t^I)^{1-\lambda} \end{aligned}$$

The Lagrangian faced by entrepreneurs reads as:

$$\mathcal{L}_t^E = w_t^P L_t^P + w_t^I L_t^I + \varrho_t \left[Y_t - \varepsilon_t^A (L_t^P)^\lambda (L_t^I)^{1-\lambda} \right] \quad (15)$$

where mc_t is the marginal cost of one unit of goods.

Which allows us to determine both wages through the marginal product of labor of each type of households:

$$L_t^P : w_t^P - \lambda \frac{Y_t}{L_t^P} = 0 \quad (16)$$

$$L_t^I : w_t^I - (1 - \lambda) \frac{Y_t}{L_t^I} = 0 \quad (17)$$

2.4 Financial intermediaries

Here, we assume perfectly competitive and frictionless financial intermediaries, which collect deposits to households and provide loans to entrepreneurs:

$$\Pi_t^B = r_t^b b_t - r_t d_t. \quad (18)$$

They face the following balance sheet constraint:

$$d_t = b_t$$

The first order condition reads as:

$$r_t^b = r_t$$

so there is no wedge between the real lending rate and the real deposit rate.

One of the key contributions of [Gerali et al. \(2010\)](#) and [Iacoviello \(2015\)](#) is to introduce imperfect financial intermediaries.

2.5 Equilibrium conditions

Aggregate consumption reads as:

$$C_t = C_t^P + C_t^I. \quad (19)$$

And total demand is given by:

$$Y_t = C_t + I_t^H, \quad (20)$$

where residential investment is given by:

$$I_t^H = H_t^I - (1 - \delta) H_{t-1}^I + H_t^P - (1 - \delta) H_{t-1}^P,$$

that is zeros because of the next equation:

$$H_t^I + H_t^P = \bar{H}.$$

ie. the endowment in dwellings is fixed, only reallocation between impatient/patient is possible.

2.6 The friction in steady state

The key friction here is the binding constraint. How are we sure that this friction is actually binding? To be sure that the friction is binding, we need to check whether in the deterministic steady state $\bar{\varphi}^I > 0$, if so then the constraint is eternally binding. First, let's remove time subscripts to Euler equations:

$$\begin{aligned}\beta_I \bar{r} - 1 &= 0 \\ 1 - \beta_I \bar{r} - \bar{\varphi}^I &= 0\end{aligned}$$

Combining these two equations through their common interest rate:

$$\bar{\varphi}^I = 1 - \frac{\beta_I}{\beta_P}$$

The constraint binds as long as $\beta_I/\beta_P < 1$, $\bar{\varphi}^I > 0$.

References

- Angelini, P., Neri, S., Panetta, F., 2011. Monetary and macroprudential policies.
- Gerali, A., Neri, S., Sessa, L., Signoretti, F. M., 2010. Credit and banking in a dsge model of the euro area. *Journal of Money, Credit and Banking* 42, 107–141.
- Guerrieri, L., Iacoviello, M., 2015. Ocabin: A toolkit for solving dynamic models with occasionally binding constraints easily. *Journal of Monetary Economics* 70, 22–38.
- Guerrieri, L., Iacoviello, M., 2017. Collateral constraints and macroeconomic asymmetries. *Journal of Monetary Economics* 90, 28–49.
- Iacoviello, M., 2005. House prices, borrowing constraints, and monetary policy in the business cycle. *American economic review* 95 (3), 739–764.
- Iacoviello, M., 2015. Financial business cycles. *Review of Economic Dynamics* 18 (1), 140–163.
- Iacoviello, M., Neri, S., 2010. Housing market spillovers: evidence from an estimated dsge model. *American Economic Journal: Macroeconomics* 2 (2), 125–64.
- Kiyotaki, N., Moore, J., 1997. Credit cycles. *Journal of political economy* 105 (2), 211–248.
- Smets, F., Wouters, R., 2003. An estimated dynamic stochastic general equilibrium model of the euro area. *Journal of the European economic association* 1 (5), 1123–1175.